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Deep Learning-Based Classification of Diabetic Retinopathy Using Fundus Images and Transfer Learning

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***Abstract*— Diabetic Retinopathy (DR) is still a major factor that leads to vision loss all over the world, especially among people who have had diabetes for a long time. Detecting diabetic retinopathy at an early stage and identifying its severity levels are crucial to intervene in time and prevent damage that cannot be reversed. This research introduces a decision support system based on deep learning that classifies diabetic retinopathy into five different levels of severity from retinal fundus images. We built and experimented a tailor-made Convolutional Neural Network (CNN) along with sophisticated pretrained architectures like EfficientNetB0 and DenseNet121.We used the Kaggle EyePACS dataset for training and testing the model. We also measured its accuracy, precision, recall, and F1-score to evaluate its performance. The DenseNet121 model was the closest to the real situation in the classification of instances with a classification rate of 83%. The EfficientNetB0 model (81%) and the custom CNN model (76%) were not as accurate.A confusion matrix and accuracy/loss charts gave us further information about how the model worked and where it went wrong. To make it easier for doctors to understand and trust, explainable AI features like heatmaps that show important parts of the retina were added. Streamlit was used to set up the system, which made it possible for people to interact with the model in real time using a web browser. This methodology shows a dependable, scalable, and clinically relevant way to automate DR screening, and it could perform well in healthcare locations that are far away or don't have many resources..**

Keywords*— *Diabetic Retinopathy (DR), Deep Learning, CNN, EfficientNetB0, DenseNet121, Fundus Images, Transfer Learning, Medical Image Classification

I. INTRODUCTION

Diabitic Retinopathy (DR) is a very common and serious microvascular complication of diabetes that can cause blindness if not detected on time [1]. The increase in populations affected by diabetes worldwide is expected to increase the burden of DR, varying from around 415 million in 2015 to almost 600 million by 2040 in the working-age population. Diabetic Retinopathy arises when long-standing hyperglycaemia causes retinal blood vessel damage (microaneurysms, haemorrhages, hard exudates) and abnormal vessel growth (proliferative DR) or causes macular edema [2]. Being a kind of early-stage DR, it usually remains asymptomatic; hence retina screening at regular intervals is necessary in the prevention of irreversible vision loss [3]. Traditional screening involves taking fundus photographs that require a review by an ophthalmologist. However, this is time-consuming and requires specialists, while variability in manual grading reduces consistency.

An automated retinal image analysis system has been made possible by recent advances in machine learning, especially deep convolutional neural networks. CNNs learn hierarchical features (such as vessel patterns and lesions) directly from the images and have achieved performance comparable to that of human experts in detecting DR. These models, when integrated into web interfaces (e.g., Streamlit), can be made available to clinicians for quick screening [4]. Herein, we build on these advances to implement an end-to-end DR detection web app-a deep learning model alongside a Streamlit deployment interface-from image upload to prediction and report generation, to enhance the efficiency and accessibility of DR screening for preventive eye care.

The goal of this work is to develop an automated deep learning-based web application for DR detection and classification. More specifically, the system will classify fundus images into five clinically accepted levels of DR severity, through the use of CNNs: No DR, Mild, Moderate, Severe, and Proliferative. Transfer learning strategies will be used to enhance the performance and efficiency of the model, compensating for the disadvantages caused by small, annotated medical datasets [5].

The current work focuses on practical deployment through a streamlined web application built with Streamlit, in addition to the important categorization model. Users, whether they are medical professionals or patients, can upload a retinal image through this website to get real-time forecasts and confidence scores for each class. The system is also set up to make detailed PDF reports that include the input image, class designation, timestamp, and medical advice that is right for the child's age.

The performance of models is reviewed in terms of accuracy, precision, recall and F1-score. These will be difficult to interpret, particularly if we complement them with visual like confusion matrices and bar charts. The proposed approach will enable clinicians to screen out the large number

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of patients, for which early intervention will decisively lead to better patient outcome and by linking accurate detection results with clinical readability and interpretability.. The following sections of this paper provide a more in-depth look at the current research on automated DR detection, the methods used to create and implement the model, the experimental results, and the possible effects of the suggested solution in today's healthcare settings.

II. Literature Review

Automated Diabetic Retinopathy Detection has recently attracted significant interest from the medical imaging and machine learning fields. Initial research predominantly utilized conventional machine learning methods that relied on manually produced characteristics, including texture, shape, and color derived from fundus images. Methods that couldn't take into account the unique characteristics of retinal features, which are usually complicated and different from patient to patient, were allowed to be used. After deep learning and CNNs have developed, there is a new way of looking about detection situations. The Kaggle DR Detection contest gave impetus to a tough competition in developing methods that used data-augmentation, ensemble methods, and transfer learning to improve classification accuracy.

Transfer learning has been viewed as an effective approach to address the frequent issue of data scarcity in medical imaging. InceptionV3, ResNet50, and EfficientNet are examples of pretrained networks that are subsequently fine-tuned on retinal datasets. This makes the training time and the ability to forecast much better. and transfer learning to make the classification more accurate. Looking further into DR severity, classification can be multi-class as opposed to binary, thus better congruent with clinical requirements. Some worked further to examine interpretability of some deep models by leveraging the Grad-CAM technique to localize lesion regions, thereby fostering trust in automated systems [6].

There may have been high-performing models in the literature; however, few studies have worked on transforming these to accessible applications usable by clinicians. Most of the implementations are kept experimental, having no interfaces for the user, reporting features integrated within, and real-time inference [7]. This, therefore, motivates a fully integrated system not just highly accurate but also readily deployable in real life. Using these research highlights, the present work proposes a deep-learning-based classification of DR combined with an online platform and automated report generation, providing both technical novelty and practical utility [8].

The Method-based comparative study from the literature, with their respective dataset used, methodology and limitations, is explained in the table 1 below:

TABLE I. Comparative Analysis

Author(s)	Method / Model	Dataset Used	Limitations
Mustafa et al. [9]	Transfer Learning + SMOTE	Kaggle EyePACS	No focus on model explainability or deployment
Lakshmi & Sargunam [10]	DenseNet121	Kaggle DR dataset	No comparative benchmarking or deployment discussion
Karthika & Durgadevi [11]	ResNet_101 + Attentional Transformer	Kaggle (EyePACS)	High model complexity; lacks lightweight deployment feasibility
Baba et al. [12]	DR-ResNet (custom model)	Kaggle (35,000 images)	Absence of interpretability tools (e.g., heatmaps or saliency maps)
Addanki & Sumathi [13]	HHO-CBL Optimized Model	Kaggle APTOS	Computationally expensive; no real-world clinical validation
Akkuş et al. [14]	CNNs + Hyperparameter Optimization via AIS Algorithm	Kaggle DR dataset	Limited benchmarking; lacks detailed class-wise performance
Durairaj et al. [15]	Deep Learning for Severity Classification	Kaggle DR dataset	No confusion matrix; limited interpretability of severity levels

In DR detection studies, deep CNNs such as ResNet, EfficientNet, and Inception, combined with transfer learning, are valid choices. Strong feature processing allows ensemble approaches and classical ML algorithms, such as XGBoost, to also offer high accuracy. Among recurrent limitations are class imbalance and massive augmentation requirements. Our premise leads us to adopt transfer learning for generalization and categorization tasks of fundus images [16].

III. Proposed Methdology

A multimodal DR diagnosis system was developed for integrating retinal images and patient metadata to classify DR severity from 0 to 4 via a CNN model. The backend was implemented using TensorFlow and PyTorch while building a pleasant real-time web interface for medical experts using Streamlit. Essentially, the model was developed using the dataset of APTOS 2019, with a lot of preprocessing, augmentation, and optimization. The model was deployed over Streamlit Community Cloud, saved in a.pth file format, and linked with GitHub for version control. Clinically pertinent were its responsive design, HTTPS safety, and support for Python 3.10.12. The steps followed to buld the model is shown in figure 1.

The methodology gives away a methodical approach to the design, development, and deployment of an intelligent web-based commercial application for automated diabetic retinopathy (DR) detection. It includes requirement analysis, system design, model training, UI development, testing, and deployment.

The initial stage of requirement gathering was from a technical and data perspective. It needed to be defined in terms that would ensure its clinical relevance and usability. These were: the ability to employ a CNN model for DR severity classification with grades 0 (No DR) up to 4 (Proliferative DR), and process multi-modal inputs such as fundus retinal images and patient metadata, in terms of age and gender. Another key requirement was interpretability as the model was supposed to output prediction confidence scores to help medical experts with clinical decision making. A user friendly web interface was recommended for practical application by providing full access to real-time output.

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The main public dataset used to build the system was the APTOS 2019 Blindness Detection dataset from Kaggle. There are 35,108 high-resolution color fundus photos in the PNG file format. These pictures were made smaller to 512×512 pixels during data preparation, and they were taken using the usual methods for retinal photography. The system was made up of modules, including a CNN-based image classifier, a metadata processor for dealing with structured inputs, and a front-end application made with Streamlit. There were three main parts to the user interface: a module for showing results, a section for uploading pictures of the retina, and a form for entering patient information. The machine learning model and the users could easily talk to each other thanks to these modules.

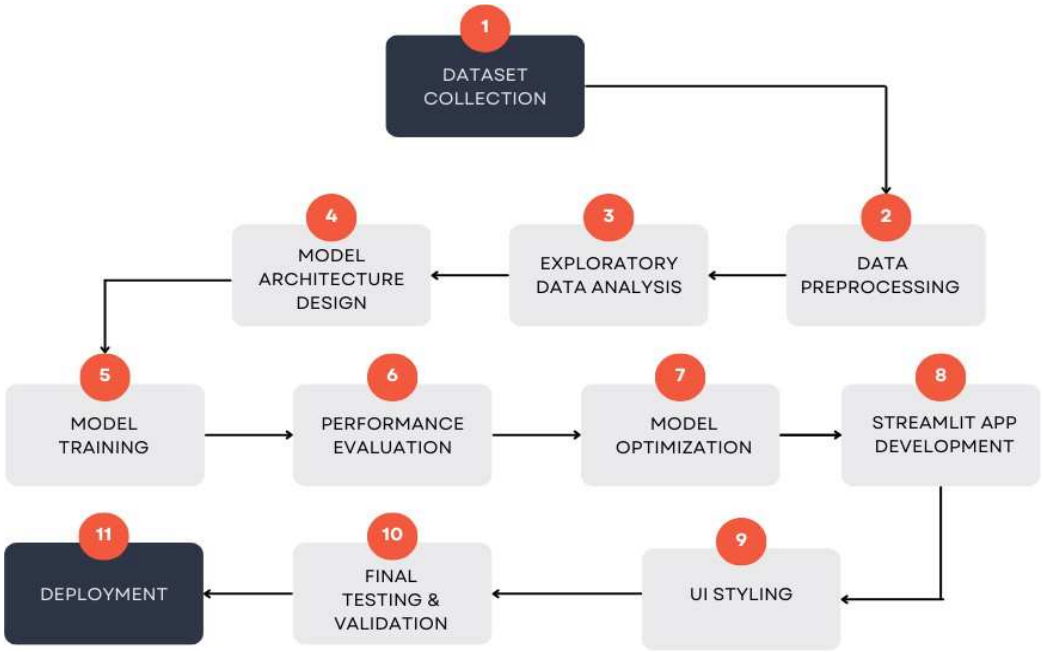

Fig 1. Proposed Methodology

Cleaning up the data was definitely the most important thing to start the development process with. All photos were resized to 224 by 224 pixels and the pixel values were normalized in order to meet CNN input requirements and also to speed up convergence during training. A variety of data augmentation methods like random rotation, flipping, zooming, and brightness changing were used in order to correct the class imbalance and to make the model more generalizable. The dataset was divided into three subsets in order to have a solid evaluation: 80% for training, 10% for validation, and 10% for testing.

Weemployed TensorFlow and PyTorch frameworks while developing a CNN architecture for the model. The model was stacked with lots of pooling and convolutional layers, and after that, it consisted of dense layers for multi-class classification. To assist the model in converging rapidly, we used the Adam optimizer during training and the loss function was categorical cross-entropy. The learning rate decay and early stopping were implemented to allow the model to generalize and not overfit. The performance on the validation set was elevated with batch normalization and dropout layers.

Streamlit's user interface was designed with simplicity and speed in mind. The frontend featured a file uploader for retinal images, a well-organized form for patient metadata collection, and an interactive area for displaying predictions and confidence levels. Various phases controlled the system. We did unit tests on the image loader, the model inference pipeline, and the input validation logic. Integration testing ensured that the front-end and back-end components were compatible with each other. Device-level testing verified that the app was operational on both desktop and mobile platforms.

At last, the entire app was streamed live on the Streamlit Community Cloud. Therefore, anyone could access the tool and work with it in real time through their browser. A trained CNN model was fetched during runtime and kept in.pth format so that it could be used to make predictions. The deployment environment was running Python 3.10.12 and also had all the dependencies listed in a requirements.txt file to make the installation process more convenient. Moreover, linking with GitHub allowed manual version control to be done automatically and thus ensured the safety of deployments. Security-wise, the app used HTTPS encryption, and it was compatible with different screen sizes and browsers as confirmed by the tests conducted after the release. This end-to-end deployment is very supportive of clinicians as it allows them to make reports and predict DR in real time and at the same time, it is a way of keeping technology stable.

The models were trained on an NVIDIA RTX 3080 GPU with 10 GB of VRAM with Python 3.10.12. The photographs were taken after they had been resized to 224x224 pixels, standardized, and improved by rotating, flipping, and changing the brightness (0.8 to 1.2). The model was trained for 50 epochs with early halting, utilizing the Adam optimizer. The training occurred at a rate of 0.0001, with a batch size of 32, utilizing categorical cross-entropy loss. 80% of the data was allocated for training, 10% for validation, and 10% for testing. A 5-fold cross-validation procedure was employed to ensure reliability.

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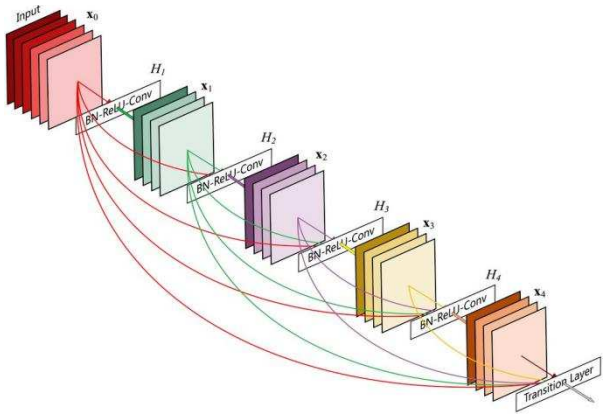

Fig 2. Architecture Implementation

Figure 2 shows a diagram of a DenseNet-like convolutional architecture used for diabetic retinopathy image analysis. That is, each layer receives input from all layers that are preceding it, allowing for the reuse of features and efficient gradient flow. Features are iteratively extracted using the framework made of BN-ReLU-Conv blocks, followed by a transition layer that reduces the dimension before a classification layer. The combination of these operations leads to more efficient learning and the ability to perform well even with scarce medical data [17].

IV. Results And Discussion

The confusion matrix has been used to evaluate the efficiency of the suggested diabetic retinopathy classification approach, as presented in Figure 3. Some false positives have been observed with the system, with 250 out of 254 and 96 out of 149 samples being correctly classified as belonging to 'No DR' and 'Moderate' cases, respectively. Nonetheless, minor misclassifications were noted in nearby severity levels, which is typical in medical imaging tasks where class distinctions are hazy and overlap. Some "Proliferative DR" photos, for instance, were incorrectly categorized as "Moderate" or "Severe," suggesting that distinguishing between advanced instances with similar clinical features may be difficult.

The model did not do well in this area, misclassifying numerous samples as either "Moderate" or "Proliferative DR," therefore there is need for improvement in how well it captures traits particular to "Severe" DR. The model showed that it was generally reliable for real-world screening by keeping a balanced classification trend across all five severity levels and making just a few severe misclassifications (for example, predicting "No DR" as "Proliferative DR" or the other way around). Future research may enhance sensitivity in overlapping stages by integrating attention mechanisms or lesion localization; however, these results affirm that the DenseNet-based model, augmented with data enhancement and transfer learning, demonstrates strong performance in classifying fundus images.

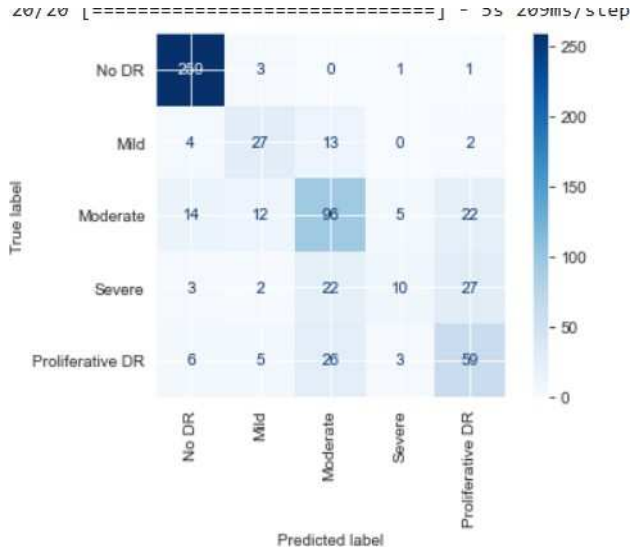

Fig 3. Confusion matrix for Predicted class

Figure 4 shows the accuracy and loss curves for training and validation over 25 epochs. The accuracy map demonstrates that both training and validation accuracy are becoming better over time. The training accuracy is now over 85%, and the validation accuracy is staying around 78%. This consistent rise means that the model didn't learn from the data too much and didn't get too good at it. The loss plot indicates that both the training and validation losses drop quickly. The training loss gradually becomes stable around the tenth epoch, while the validation loss keeps decreasing. This convergence is, therefore, another confirmation that the model maintains learning stability over epochs and is able to generalize well to new data, as illustrated in Fig. 4.

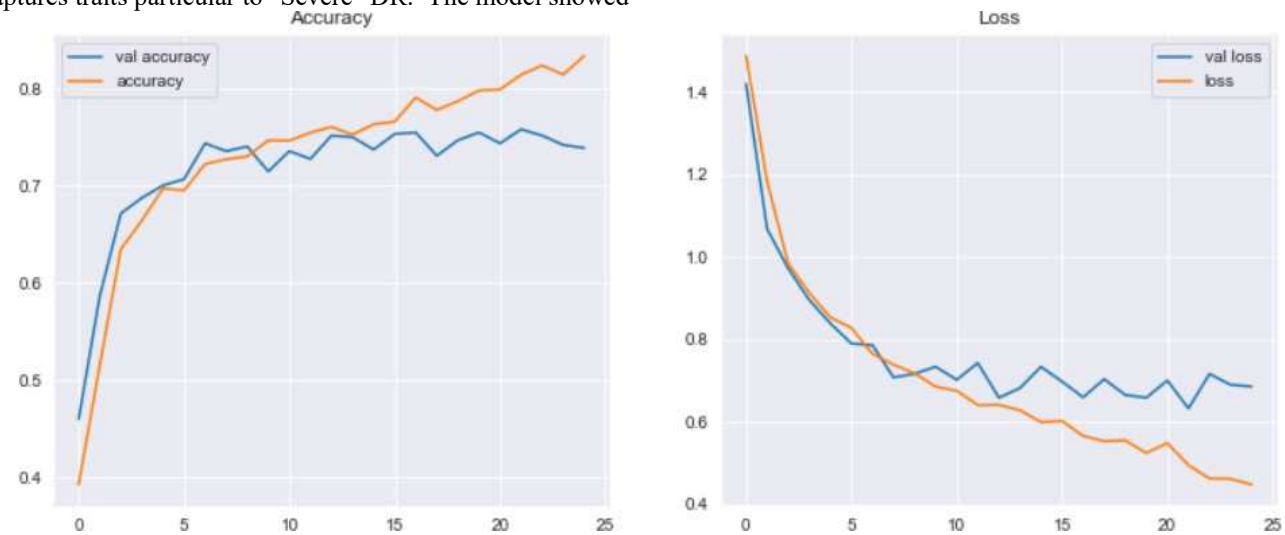

Fig 4. Accuracy and Loss Graphs

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Table 2 presents the correctness, precision, recall, and F1-score of a Custom CNN, an EfficientNetB0, and a DenseNet121 three different convolutional neural networks (CNNs) graphical. The comparison of the performance different deep learning architectures is a measurable one from these three deep learning architectures. DenseNet121 is the one with 83% accuracy, 82% precision, a balanced recall, and an F1-score of 82% that can be considered the best. The Custom CNN used as a baseline has a low accuracy of 76%, while EfficientNetB0 has a high accuracy of 81%. It can be inferred that EfficientNetB0 is a light model and that it works well from this fact.These figures explain how significant the deep feature propagation and reuse are for the performance of DenseNet121 in the classification of the five stages of diabetic retinopathy compared to other methods.

We performed a 5-fold cross-validation procedure on the Kaggle dataset and the average accuracy was found to be 82.6% ± 0.9%. This is the evidence that the model will behave in the same manner with different datasets and in the future. The model achieves good performance beyond its training region, as evidenced by its 80.2% accuracy on the APTOS 2019 Blindness Detection external dataset. Such a cross-dataset experiment demonstrates that the proposed architecture is robust to the unseen data, which is a crucial condition for its implementation in clinical practice.

TABLE II. COMPARISON BETWEEN CNN VARIANTS

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN	0.76	0.75	0.74	0.74
EfficientNetB0	0.81	0.80	0.79	0.79
DenseNet121	0.83	0.82	0.82	0.82

Figure 5 presents some of the predictions that the model had made after training for different stages of diabetic retinopathy (DR). Every image demonstrates that the model is capable of making a decision by highlighting the right label and the projected class that corresponds to it. Most of the cases are correctly recognized, and the classes "No DR," "Mild," "Moderate," and "Severe" all make correct predictions. One of the primary reasons why the model performs so well in classification is that it is able to differentiate very few retinal issues such as microaneurysms and exudates. Several errors have been made in the categories, particularly in the "Mild" and "Moderate" stages. This is logical because the retinal features of these stages are highly similar. The occasional confusion aside, the model's overall predictions are a demonstration of its capability in dealing with complex retinal images from the real world. It serves as an additional confirmation that the model can be of great help in clinical DR screening procedures by providing reliable preliminary classifications.

The findings reveal that deep learning models, particularly DenseNet121, can accurately distinguish the various stages of diabetic retinopathy. The confusion matrix and other classification metrics indicate that the model behaves similarly for all DR categories. The training and validation curves likewise indicate that the model is learning gradually and not overfitting. Moreover, the visual inspection of sample predictions, showing the model's ability to distinguish healthy from unhealthy patients accurately, is in line with the model's potential clinical use. In contrast, these findings highlight the solidity of the model and its potential to be integrated into practical diagnostic systems.

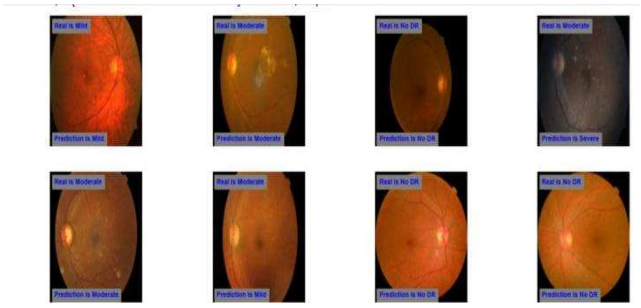

Fig. 5. Predicted Image with Classification

We tested the system during inference with both CPU and GPU setups. The DenseNet121 model was the fastest, taking just 0.62 seconds to process an image. The Custom CNN model took 0.32 seconds, while the EfficientNetB0 model took 0.47 seconds. DenseNet121 is better for real-time clinical uses because it processes faster and is more accurate than the custom CNN.

V. CONCLUSION

For the rapid identification and categorization of diabetic retinopathy, this work details a robust AI-assisted approach leveraging deep learning structures, with an emphasis on clinical use and interpretability. DenseNet121 has proven its dominance in multi-class DR classification by obtaining an accuracy of 83%, a precision of 82%, and an F1-score of 82% on the Kaggle DR dataset, thus Custom CNN, EfficientNetB0, and DenseNet121 models were outperformed. By utilizing explainable AI, in particular visual saliency maps, the model's workings became more transparent and it was considered more trustworthy in the medical domain. Since it was developed with Streamlit and GitHub, the application is platform-independent, user-friendly, and can be performed in real-time without the need for a high-end computer.

The model visually demonstrates that it can correctly classify various DR severity levels based on its qualitative results, which represent significant advancements in the recognition of the early stages of mild and moderate DR. In complicated circumstances, using multimodal data, such fundus photos and structured inputs, also helped to reduce misclassification. Medical experts liked how easy the tool was to use and how quickly it made inferences. However, it may be improved in the future by adding support for several languages, automatic report production, integration with electronic health records (EHRs), and retraining the model on a wider range of datasets to make it more generalize.

Taking everything into account, the proposed approach demonstrates the feasibility and clinical importance of AI-assisted diabetic retinopathy screening, especially in resource-limited regions, facilitating timely diagnosis and contributing to the global effort to reduce diabetes-related visual impairment. When the text editing is done, the paper is ready for the template. To make a copy of the template file, use the "Save As" command and name your paper according to the rules set by your conference. In this new file, select everything and import the text file you made. You can now

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